

Day 21 What is Recursion?

This is the best resource to learn recursion completely
Youtube Video

<https://youtu.be/OvTH-7ESoRA?si=6sRXjD0vgdbX-AHZ>

Recursion is when a function calls itself to solve a problem by breaking it down into smaller, similar sub-problems.

The Core Idea

Instead of solving a big problem all at once, recursion solves it by:

1. Solving a tiny piece of the problem
2. Calling itself to solve the rest (which is a smaller version of the original problem)
3. Repeating until it reaches the simplest case (called the **base case**)

Key Components

1. Base Case - The stopping condition. Without this, the function would call itself forever.

2. Recursive Case - Where the function calls itself with a simpler version of the problem.

recursion is used in Different Types of Data Structures like

- 1. Tree Problems (Most Common)**
- 2. Graph Problems**
- 3. Backtracking Problems (Very Important)**
- 4. Divide and Conquer**
- 5. Dynamic Programming (with Memoization)**
- 6. String Problems**
- 7. Array Problems**
- 8. Linked List Problems**
- 9. Mathematical Problems**

10. Game Theory & Puzzles

so please learn this topic and practice problems

Basic Practice Problems To master recursion

- ❑ Factorial of a number
- ❑ Fibonacci number (nth term)
- ❑ Sum of digits of a number
- ❑ Reverse a number
- ❑ Check palindrome number
- ❑ Power of a number (x^n)
- ❑ GCD of two numbers
- ❑ Print array elements using recursion
- ❑ Find maximum element in an array
- ❑ Tower of Hanoi